WASP-43b

R-0335

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_260213` · created 2026-07-18T05:34:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461084.9014 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.057
Transit depth (Rp/Rs)^2	3.05 +/- 1.97 [%]
Orbital Inclination (inc)	86.66 +/- 2.91
Airmass coefficient 1 (a1)	1.75 +/- 0.53
Airmass coefficient 2 (a2)	-0.44 +/- 0.3
Scatter in the residuals of the lightcurve fit is	25.92 %
Best Comparison Star	#1 - [474, 217]
Optimal Aperture	3.24
Optimal Annulus	11.2
Transit Duration (day)	0.0594 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.057 vs 0.1594 (deviation 0.27σ)
Duration fit vs published	0.0594 d vs 0.0512 d (deviation 1.48σ)
Mid-time O–C	7.0 min
Depth SNR	3.1
Residual scatter	25.92 %

Light curve

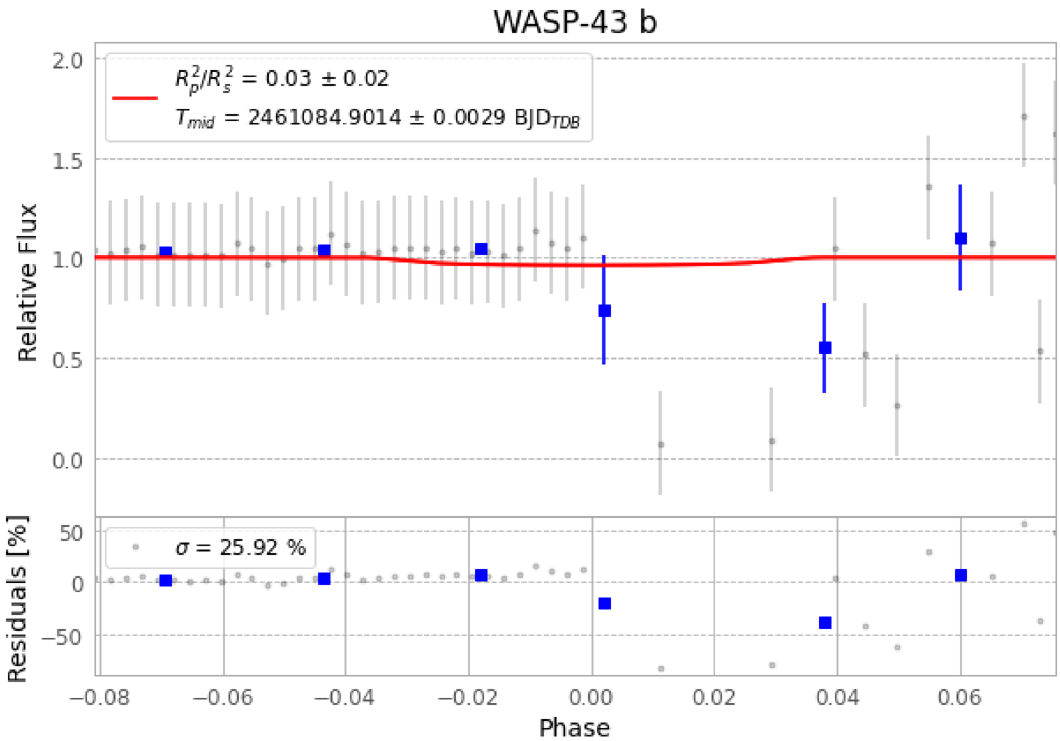

Figure 1 — FinalLightCurve_WASP-43 b_2026-02-13.png (SHA-256 61d2ee3860530989…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 170], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5458e55514313f35…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 061fc91`

Data provenance

62 FITS frames (41 MB), `WASP-43260213075415.FITS` → `WASP-43260213105715.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260213.log (b9c45776c74a…), FinalLightCurve_WASP-43 b_2026-02-13.png (61d2ee386053…), FinalLightCurve_WASP-43 b_2026-02-13.csv (2d75c498a9ec…)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)